

are able to correctly calibrate their test to account for the introduced dependencies. This framework requires asymptotic normality of the test statistic under the null hypothesis, however, and hence they are by design restricted to using the linear-time MMD estimate. We observe in our experiments that using this estimate results in a significant loss in power when compared to tests which use the quadratic-time statistic. Yamada et al. (2019) also use post-selection inference to obtain a feature selection method based on the MMD, where the chosen features best distinguish the samples.

In a different setting, Wynne and Duncan (2022) study the efficiency of MMD-based tests when dimension increases, and propose an MMD-based two-sample test for Functional Data Analysis, a framework in which the samples consist of functions rather than of data points. In this setting, Wynne and Nagy (2021) study the connections between kernel mean embeddings and statistical depth (i.e. how representative a point is from a given measure).

5 Experiments

For our aggregated MMDAgg test, we first introduce in Section 5.1 four weighting strategies and a family of collections of bandwidths motivated by Corollary 10. Those collections depend on some parameters which would usually need to be chosen by the user, by contrast, we introduce in Section 5.2 a parameter-free adaptive collection for MMDAgg, which we recommend using in practice. We then present in Section 5.3 some other state-of-the-art MMD-based two-sample tests we will compare ours to. In Section 5.4, we provide details about our experimental procedure. We show that our aggregated MMDAgg test obtains high power on both synthetic and real-world datasets in Sections 5.5 and 5.6, respectively. In Section 5.7, we observe that MMDAgg retains power even in the continuous limit of the collection of bandwidths. We show in Section 5.8 that, on image shift experiments, MMDAgg matches the power of tests using neural networks, even for large sample sizes. Finally, in Section 5.9, we briefly report the results from the additional experiments presented in Appendix A.

5.1 Weighting strategies and fixed bandwidth collections for MMDAgg

The positive weights $(w_\lambda)_{\lambda \in \Lambda}$ for the collection of bandwidths Λ are required to satisfy $\sum_{\lambda \in \Lambda} w_\lambda \leq 1$. As noted in Equation (14), rescaling all the weights to ensure that $\sum_{\lambda \in \Lambda} w_\lambda = 1$ does not change the output of our aggregated test $\Delta_\alpha^{\Lambda^w, B_{1:3}}(\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}, \mathbb{Z}_{B_2})$. For this reason, we propose weighting strategies for which $\sum_{\lambda \in \Lambda} w_\lambda = 1$ holds.

For any collection Λ of N bandwidths, one can use *uniform* weights which we define as

$$w_\lambda^u := \frac{1}{N} \quad \text{for } \lambda \in \Lambda.$$

Using uniform weights should be prioritised if the user does not have any useful prior information to incorporate in the test. The choice of weights is entirely up to the user; the weights can be designed to reflect any given prior belief about the location of the ‘best’ bandwidths in the collection. Nonetheless, we also present three standard weighting strategies for incorporating prior knowledge when dealing with a more structured collection of bandwidths.

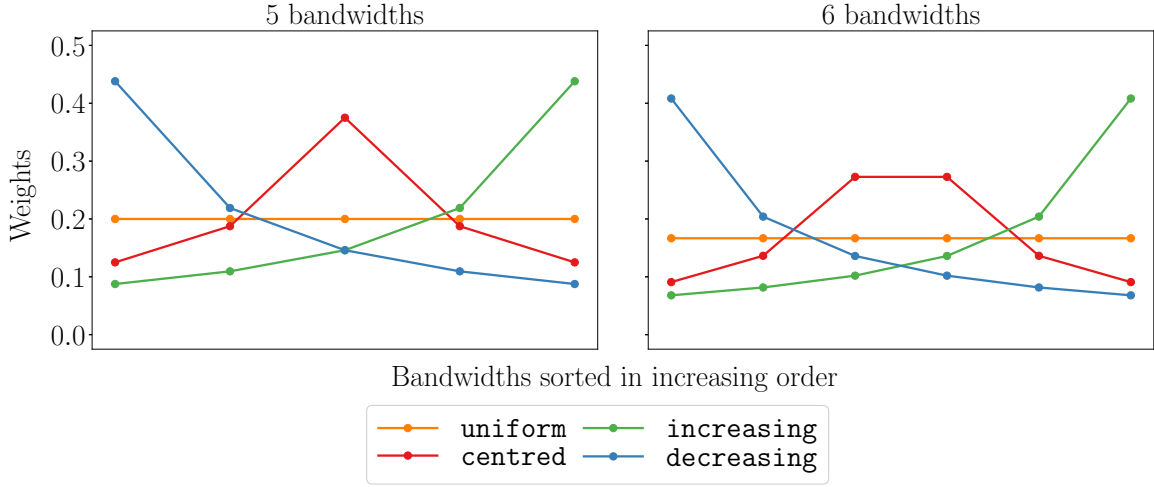


Figure 1: Weighting strategies.

Consider the case where we have some reference bandwidth $\lambda_{ref} \in (0, \infty)^d$ and we are interested in aggregating scaled versions of it, that is, we have an ordered collection of N bandwidths defined as $\lambda^{(i)} := c_i \lambda_{ref}$, $i = 1, \dots, N$, for positive constants $c_1 < \dots < c_N$. If we have no prior knowledge, then we would simply use the aforementioned uniform weights. Suppose we believe that, if the two distributions differ, then this difference would be better captured by the smaller bandwidths in our collection, in that case we would use *decreasing* weights

$$w_{\lambda^{(i)}}^d := \frac{1}{i} \left(\sum_{\ell=1}^N \ell^{-1} \right)^{-1} \quad \text{for } i = 1, \dots, N.$$

On the contrary, if we think that the larger bandwidths in our collection are well suited to capture the difference between the two distributions, if it exists, then we would use *increasing* weights

$$w_{\lambda^{(i)}}^i := \frac{1}{N+1-i} \left(\sum_{\ell=1}^N \ell^{-1} \right)^{-1} \quad \text{for } i = 1, \dots, N.$$

Finally, if our prior knowledge is that the bandwidths in the middle of our ordered collection are the most likely to detect the potential difference between the two densities, then we would use *centred* weights which, for N odd, are defined as

$$w_{\lambda^{(i)}}^c := \frac{1}{\left| \frac{N+1}{2} - i \right| + 1} \left(\sum_{\ell=1}^N \left(\left| \frac{N+1}{2} - \ell \right| + 1 \right)^{-1} \right)^{-1} \quad \text{for } i = 1, \dots, N,$$

and, for N even, as

$$w_{\lambda^{(i)}}^c := \frac{1}{\left| \frac{N+1}{2} - i \right| + \frac{1}{2}} \left(\sum_{\ell=1}^N \left(\left| \frac{N+1}{2} - \ell \right| + \frac{1}{2} \right)^{-1} \right)^{-1} \quad \text{for } i = 1, \dots, N.$$

All those weighting strategies are inspired from the weights of Corollary 10 which are defined as $w_{\lambda(i)} := i^{-2}(\sum_{\ell=1}^{\infty} \ell^{-2})^{-1}$ for $i \in \mathbb{N} \setminus \{0\}$, where the square exponent is required in order to have a convergent series. However, in practice, using square exponents in our weights would assign extremely small weights to some of the bandwidths in our collection. This would be almost equivalent to disregarding those bandwidths in our aggregated MMDAgg test, which is not a desired property since if we are not interested in testing some bandwidths, then we would simply not include them in our collection. For this reason, we have defined our weighting strategies without the square exponent. We provide visualisations of our four weighting strategies for collections of 5 and 6 bandwidths in Figure 1.

In our experiments, we refer to our aggregated MMDAgg test with those four weighting strategies as: **MMDAgg uniform**, **MMDAgg decreasing**, **MMDAgg increasing** and **MMDAgg centred**. For these, we use $B_1 = 500$ simulated test statistics to estimate the quantiles, $B_2 = 500$ simulated test statistics to estimate the probability in Equation (13) for the level correction, and $B_3 = 100$ iterations for the bisection method. Motivated by Corollary 10, those tests are used with collections of bandwidths of the form

$$\Lambda(\ell_-, \ell_+) := \left\{ 2^\ell \lambda_{med} \in (0, \infty)^d : \ell \in \{\ell_-, \dots, \ell_+\} \right\} \quad (15)$$

for $\ell_-, \ell_+ \in \mathbb{Z}$ such that $\ell_- < \ell_+$, where the median bandwidth λ_{med} is

$$(\lambda_{med})_i := \text{median}\{|w_i - w'_i| : w, w' \in \mathbb{X}_m \cup \mathbb{Y}_n, w \neq w'\}$$

for $i = 1, \dots, d$. We note that, for each experiment, we have chosen the values ℓ_- and ℓ_+ which highlight the differences between the four weighting strategies. In practice, it is not clear how to choose those values, instead, we recommend using the adaptive parameter-free collection of bandwidths introduced in Section 5.2 with uniform weights.

5.2 Adaptive parameter-free collection of bandwidths for MMDAgg

For radial basis function kernels (*i.e.* kernels $k(x, y)$ which can be written as a function of $\|x - y\|$ for some norm $\|\cdot\|$), we recommend using a collection of bandwidths which, intuitively, discretises the interval between the smallest and the largest of the inter-sample distances⁷

$$D := \{\|x - y\| : x \in \mathbb{X}_m, y \in \mathbb{Y}_n\}.$$

More formally, we use

$$\Lambda = \left\{ \left(\frac{2\lambda_{max}}{\lambda_{min}/2} \right)^{(i-1)/(N-1)} \lambda_{min}/2 : i = 1, \dots, N \right\}$$

which is a discretisation of the interval $[\lambda_{min}/2, 2\lambda_{max}]$ using N points. We let λ_{min} be the minimum value in D . If this value is smaller than 0.1, we instead use the 5% smallest value in D for λ_{min} , if this quantity is still smaller than 0.1, we use $\lambda_{min} = 0.1$. For λ_{max} , we use the maximum value of D or 0.3 if this maximum is smaller than 0.3. In practice, we recommend using $N = 10$ points, as can be observed in Figure 6 the power remain the same when increasing N to be larger (*i.e.* $N = 100$ or $N = 1000$). Since in general we might not

7. In practice, D can be computed using at most 500 samples from $\mathbb{X}_m$ and 500 samples from $\mathbb{Y}_n$.